A Randomized Trial of Alcohol Telemedicine in Primary Care: Pharmacotherapy and Referral Outcomes

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ABSTRACT

IMPORTANCE: Unhealthy alcohol use is commonly identified in primary care, yet most patients do not receive any treatment services. Pharmacists can complement the work of primary care physicians (PCPs), but it is unknown whether they increase patient access to services for alcohol use problems.

OBJECTIVE: To examine implementation outcomes of a primary care–based, pharmacist-delivered service for patients with unhealthy alcohol use.

DESIGN: A pragmatic, cluster-randomized trial conducted in a large healthcare system between 9/1/2021 and 8/31/2023.

SETTING: Primary care clinics block-randomized by facility to an intervention arm ($N=8$) or Usual Care ($N=8$).

PARTICIPANTS: Primary care providers (204 intervention arm, 132 Usual Care arm) from study clinics.

INTERVENTION: Alcohol Telemedicine Consultation (ATC), involving telephone contact, assessment, motivational enhancement, and specialty referral facilitation.

MAIN OUTCOMES AND MEASURES: Prescriptions for an FDA-approved alcohol use disorder (AUD) medication and referrals to specialty treatment for individuals (age 18+) who screened positive for unhealthy alcohol use or received an AUD diagnosis between 9/1/2021 and 8/31/2023 summarized by arm.

RESULTS: *Bivariate:* Collectively, ATC arm providers made 161 referrals to ATC in year 1 and 126 referrals in year 2. Usual Care arm providers made more referrals to specialty treatment (mean (SE) 3.14 (0.28) vs. 2.34 (0.20) in year 1 and 3.24 (0.27) vs.2.43 (0.21) in year 2).

MULTIVARIABLE: Multivariable analyses using zero-inflated Poisson models adjusting for provider characteristics showed that ATC-arm providers prescribed 66% more AUD medications (RR=1.66, $p=0.001$) and made 15% more referrals to specialty treatment (RR=1.15, $p=0.049$) in year 2 compared to Usual Care.

CONCLUSIONS AND RELEVANCE: Referral to ATC may offer an effective approach for improving treatment utilization by increasing pharmacotherapy in primary care, but providers may need training boosters over time.

TRIAL REGISTRATION: ClinicalTrials.gov # 1564446.

Key Points

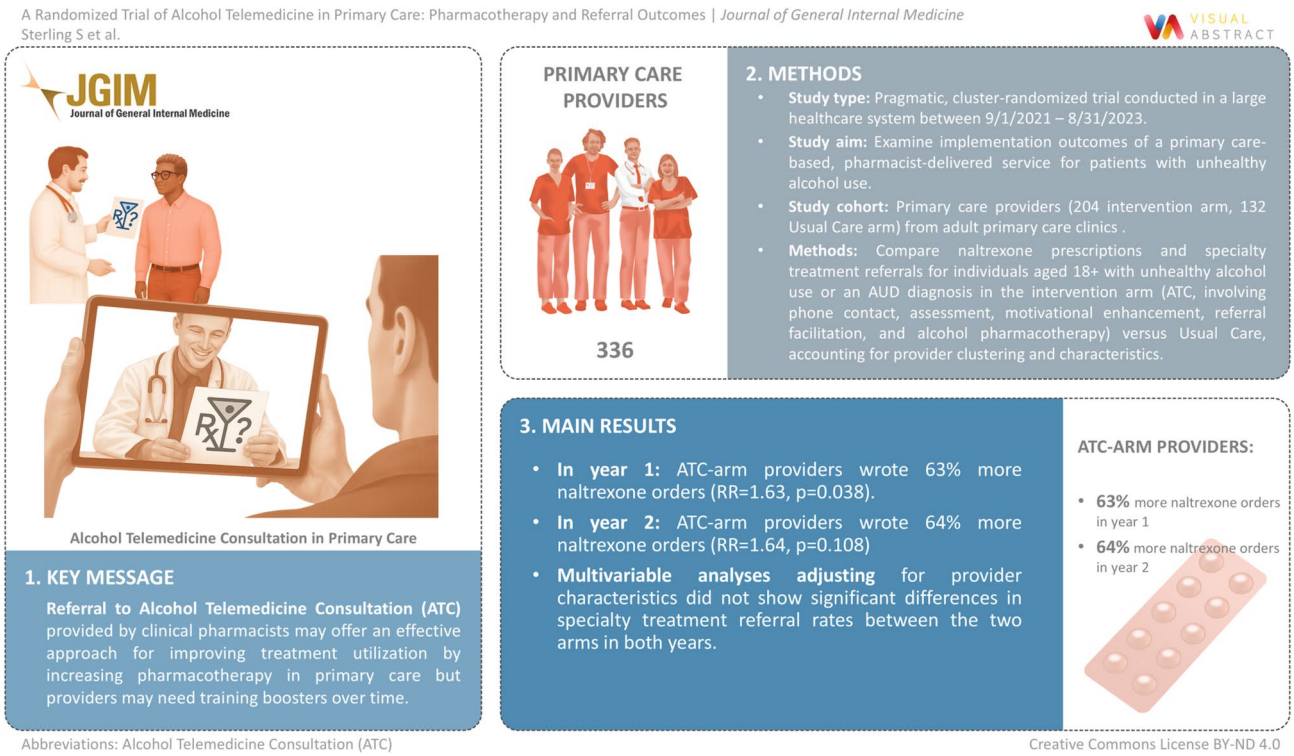
Question: Does a pharmacist-delivered telehealth intervention increase alcohol treatment referrals and alcohol use disorder (AUD) medications prescribing among primary care physicians?

Findings: In this pragmatic, cluster-randomized clinical trial that included 336 providers, the intervention arm had 63% more AUD medication orders in year 1, and 65% more in year 2.

Meaning: This telehealth intervention may be an effective approach for improving treatment utilization by increasing alcohol pharmacotherapy prescribing and specialty treatment referral in primary care.

Received September 19, 2024
Accepted June 25, 2025
Published online July 9, 2025

Graphical Abstract



J Gen Intern Med 41(1):6–15
DOI: 10.1007/s11606-025-09708-y
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INTRODUCTION

Unhealthy alcohol use poses significant challenges to healthcare providers and patients and can adversely affect acute and chronic medical conditions,[1] impede patients from seeking preventive healthcare,[2] and result in avoidable emergency and inpatient care.[3] Unhealthy alcohol use is also associated with alcohol use disorder (AUD) and other substance use disorders,[4] including opioid use disorder.[5]

Alcohol treatment, both psychosocial and pharmacotherapeutic, has demonstrated efficacy and effectiveness, yet few people who need it receive it.[6,7] Reasons include availability, logistical barriers,[8] and stigma.[9] Primary care–based intervention for unhealthy alcohol use[10] could expand treatment options and prevent progression to AUD.

Primary care is the main — sometimes the only — point of contact with the health system, making it an ideal setting for treatment. The most notable effort to manage unhealthy alcohol use in primary care is Screening, Brief Intervention, and Referral to Treatment (SBIRT).[11,12] Brief interventions can help patients reduce unhealthy alcohol use[13,14] but are often insufficient for more severe problems.[15] Referral to specialty alcohol treatment is part of the SBIRT model, yet brief interventions have not increased receipt of treatment, even when readily available.[6,16,17]

Medications to treat AUD have long been available but underused.[18] The three FDA-approved drugs for AUD are naltrexone, disulfiram, and acamprosate, with other medications (e.g., gabapentin, topiramate) prescribed off label.[19] Robust evidence supports their safety and efficacy, particularly naltrexone, in alcohol use reduction,[7,20–27] including among those with mild-to-moderate AUD.[28,29] However, these medications are rarely prescribed, particularly in primary care.[30–32] Barriers include lack of knowledge, confidence in prescribing,[33] limited appointment time, and competing priorities.[34,35]

Pharmacists have become key resources for extending and complementing the work of primary care physicians (PCPs)[36] as clinical demands have increased.[37,38] Pharmacists have expanded scopes of practice in 43 states[39] and perform important functions, including assessments; ordering and interpreting laboratory tests; referrals to other clinicians; prescribing and managing medications under protocol; and collaborating to manage patient health.[40] They increasingly receive training in Motivational Interviewing (MI),[41] leading to improved medication adherence and patient satisfaction.[42] Research indicates that pharmacists are aware of the adverse impact of alcohol on health[43] and may be well suited to treat AUD in primary care, but it is unknown whether and how they might increase patient access to pharmacotherapy and other services for alcohol problems.

To address this evidence gap, we conducted a pragmatic, cluster-randomized trial in 16 adult primary care clinics, to examine the effectiveness of a multi-faceted intervention — the Addiction Telemedicine Consultation service ("ATC") — that provided flexible pharmacy consultation to PCPs and their patients with unhealthy alcohol use.[44] ATC was informed by pilot work using a consultation-based model with Addiction Medicine specialists[45] and a large-scale study of a pharmacist-delivered, in-person Academic Detailing Program in the Veterans Health Administration (VHA) which found more use of alcohol pharmacotherapy post-implementation.[46] We compared ATC and Usual Care (UC) arms on alcohol pharmacotherapy prescribing rates and specialty Addiction Medicine treatment (hereafter called "specialty treatment") referral rates over 2 years. We hypothesized that the ATC arm would have higher prescribing and treatment referral rates than UC.

METHODS

Study Setting

Kaiser Permanente Northern California (KPNC) is a non-profit health system of ~4.5 million members, about one-third of Northern California's population, with demographics similar to the statewide population.[47] Its membership is socio-economically diverse and representative of the US population with access to care.[47] KPNC offers coverage via employer-based plans, Medicare, Medicaid, and insurance exchanges, and provides psychiatry and Addiction Medicine treatment as a covered benefit. Specialty services in this medical system include detoxification, intensive outpatient treatment for individuals with severe AUD, and harm reduction and motivational ("contemplation") groups for individuals considering making a change to their drinking. PCPs are encouraged to refer people with likely mild or moderate AUD to specialty treatment for evaluation, diagnosis, and motivational intervention — consistent with national guidelines for essential services in specialty treatment.[48]

Informed by the SBIRT literature, including work by this research team,[49] in 2013 KPNC implemented systematic alcohol SBIRT in adult primary care using an NIAAA evidence-based screening instrument embedded in the electronic health record (EHR), i.e., "How many times in the past three months have you had 5 or more drinks containing alcohol in a day?" for men aged 18–65 years, or "4 or more drinks" for men ≥ 66 years and women ≥ 18 years, followed by two questions about typical drinking days per week and typical number of drinks per day.[10] This standardized SBIRT, which includes medical assistant training in screening and physician training in brief intervention and referral to treatment,[50] served as the UC condition.

The Alcohol Telemedicine Consultation (ATC) Intervention

The ATC intervention was designed to be an efficient, centralized approach whereby pharmacists treated primary care patients with unhealthy alcohol use. The intervention and study design have been described in the protocol.[44] PCPs in the ATC arm accessed centrally located ATC pharmacists using an electronic referral mechanism in the EHR. The ATC pharmacists were two experienced ambulatory care practitioners who received additional training in patient assessment, MI, non-stigmatizing and trauma-informed approaches to discussing alcohol use, referral facilitation, and alcohol pharmacotherapy. Training occurred biweekly over 8 months, for a total of 26 h, with 1-h weekly consultation post-launch, as needed. The service included direct patient contact via video or telephone, assessment, psychoeducation, motivational enhancement, facilitation of patient engagement in treatment, and pharmacotherapy prescribing. The pharmacists provided pharmacotherapy and specialty treatment referrals — based on clinical assessment and patient preference.

Pharmacists prescribed oral naltrexone as the first-line medication, per prescribing guidelines,[51] robust efficacy evidence,[52] once-daily dosing, favorable safety profile, and limited side effects. The protocol presented acamprosate as an alternative for patients with hepatic dysfunction or opioid use[53]. Intervention-arm PCPs were provided with an hour-long lunchtime training and brief informational boosters on unhealthy alcohol use, pharmacotherapy, and how to access the ATC service.

Study Design and Sample

All adult primary care clinics ($n = 16$) in each of two large, urban medical centers were block-randomized by facility to ATC ($n = 8$ clinics) or UC ($n = 8$ clinics). The study sample consisted of providers in the ATC arm ($n = 204$ providers) and UC arm ($n = 132$). This study was reviewed and approved by the Kaiser Permanente Northern California Institutional Review Board.

Data Sources and Measures

We extracted alcohol pharmacotherapy prescription orders and specialty treatment referral rates from KPNC's EHR system by providers for individuals (age 18 +) who screened positive for unhealthy alcohol use via routine SBIRT or received an AUD diagnosis (*all F10.xxx excluding remission, G31.2xx, G62.1xx, G72.1xx, I42.6, K29.2, K70.xx, K86.0*), between 9/1/2021 and 8/31/2023 ($N = 38{,}132$ individuals), which integrates data on services use with patient enrollment, demographics, utilization including medications, ICD-10 codes,[54] procedures, laboratory results, and vital signs. Inpatient and outpatient prescription data — including

National Drug Codes and standard drug class codes, and diagnosis codes associated with medication orders — are captured for nearly 100% of enrollees.[55] PCP-level characteristics (age, sex, race/ethnicity, panel size, specialty [internal medicine vs. other (family practice or general medicine)], and length of service at KPNC (proxy for total years of experience, which was not available) were used as covariates in multivariable models.

Alcohol Pharmacotherapy Prescribing. The pharmacotherapy prescribing rate was calculated as the number of orders divided by the total number of eligible patients in the PCP's panel (i.e., patients who had a positive alcohol screening, brief intervention for unhealthy alcohol use, or AUD diagnosis). The prescribing rate was calculated for each 12-month period (9/1/21–8/31/22 and 9/1/22–8/31/23), to allow for examination of practice changes over time. Orders issued by the ATC pharmacists were assigned to the PCP who initiated the referral. As prespecified in the protocol and ClinicalTrials.gov registration (#1,564,446), we examined all approved AUD medications (naltrexone, acamprosate, and disulfiram) together as the primary prescribing outcome among the full sample of eligible patients, including both those with AUD diagnoses and those who reported unhealthy alcohol use. All patients referred to ATC for alcohol use problems, who were deemed by their PCP to require more than brief advice for their alcohol use, were considered potentially eligible for pharmacotherapy. This approach was developed in consultation with subject matter experts and NIH officials who indicated interest in expanding the use of pharmacotherapy to a broader spectrum of patients who might benefit from these medications. This was also supported by prior work which examined the use of pharmacotherapy for people with mild-to-moderate AUD.[28,29] In this health system, PCPs are encouraged to refer patients to specialty substance use treatment for thorough assessments rather than diagnosing AUD in primary care themselves, unless the diagnosis is abundantly clear. For ATC, intervention-arm PCPs were trained to provide usual care (i.e., brief advice) to patients who they deemed unlikely to meet criteria for AUD but were encouraged to refer patients to ATC those patients who they felt might meet criteria for AUD, need more than brief advice, and/or might benefit from specialty treatment but declined it or were ineligible or inappropriate for specialty treatment.

Referrals to Care for Unhealthy Alcohol Use. We extracted from the EHR all PCP-initiated referrals for unhealthy alcohol use during the study period and created separate indicators for referrals made to ATC and referrals made to specialty treatment. Referral rates for specialty treatment were calculated as the number of referrals for each 12-month period in each arm divided by the total number of providers in that arm, to allow for examination of practice changes over time.

Data Analysis

We examined bivariate differences in implementation outcomes (defined as AUD pharmacotherapy prescribing and specialty treatment referrals) between the ATC and UC arms annually to understand differences between the arms and changes during the intervention period. Eligible individuals in each year were included in both years. Preliminary models examined the relationship between study arm and the primary implementation outcomes by provider. To enhance our understanding of this relationship, we then estimated multivariable models including provider-level characteristics (age, gender, race/ethnicity, specialty, and length of service at KPNC). Since many providers did not prescribe any AUD medication and the distribution of orders is a count variable with excess zeroes, we examined differences in annual prescribing and specialty treatment referral rates between PCPs in the two arms using a zero-inflated Poisson (ZIP) model[56] with panel size as an offset and accounted for nesting of providers within clinics. The ZIP model splits the outcome into two components: a logistic regression that models the likelihood of the outcome (e.g., whether a provider wrote a

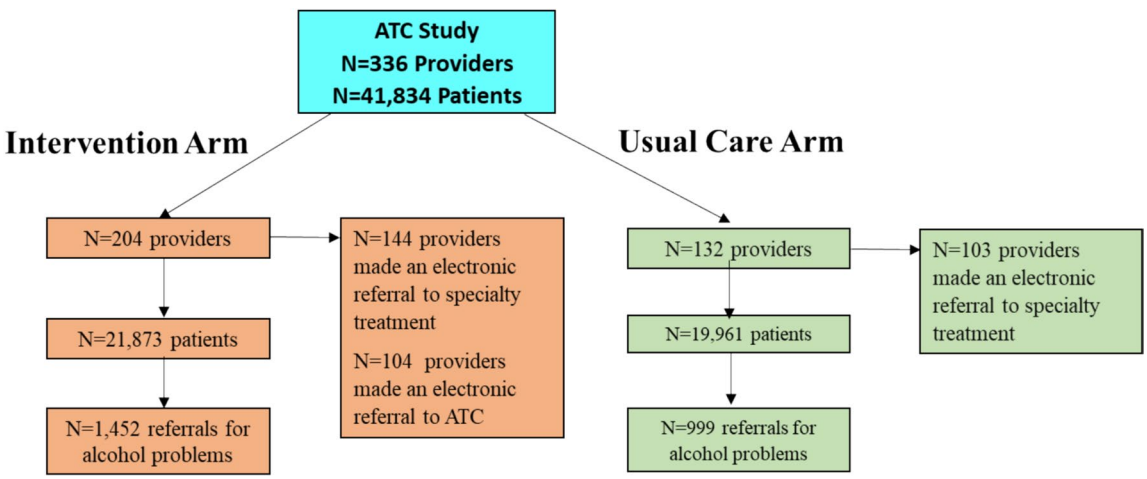

Figure 1 Addiction Telemedicine Consultation (ATC) study at a glance

Table 1 Characteristics by Treatment Arm

	ATC arm	Control arm	
Provider characteristics	**(*N* = 204)**	**(*N* = 132)**	***p*-value**
Gender			
No. of male (%)	93 (45.6)	44 (33.3)	0.026
Age (std. dev)	36.5 (9.4)	40.3 (9.7)	< 0.001
Race			
No. of Asian (%)	109 (53.4)	80 (60.6)	0.350
No. of Black (%)	15 (7.4)	5 (3.8)	
No. of White (%)	60 (29.4)	38 (28.8)	
No. of other/unknown (%)	20 (9.8)	9 (6.8)	
Ethnicity			
Hispanic (%)	16 (7.8)	6 (4.5)	0.233
Specialty			
Internal Medicine (%)	141 (69.1)	88 (66.7)	0.637
Years at KP (std. dev)	6.6 (7.7)	9.4 (8.9)	0.003

Eligible patients screened positive for unhealthy alcohol use, received a brief intervention as documented by ICD-10 alcohol counseling Z-code, received an AUD diagnosis, and/or received a referral for an alcohol problem

prescription for an AUD medication) and a second part that examines the count of the outcome (e.g., number of AUD medication prescriptions). The key covariate of interest in these models was the indicator variable of treatment arm, and the exponent of the coefficient represented the rate ratio for the ATC arm vis-à-vis the UC arm. Statistical significance was defined at $p < 0.05$ and all analyses were conducted using SAS®.

RESULTS

There were 336 PCPs in the study (ATC arm = 204, UC arm = 132). At least 25% of intervention-arm providers made a referral to the ATC service ($n = 50$ and 62 for years 1 and 2, respectively) and 75% made a referral to specialty treatment services (Fig. 1) in each year; 75% and 81% (99 and 107) of UC providers made a referral to specialty treatment in years 1 and 2, respectively (Fig. 1). More providers in the ATC arm were male (45.6% vs. 33.3%; $p = 0.026$), were younger (36.5 vs. 40.3 years; $p < 0.001$), and had fewer years of service at KPNC (6.6 vs. 9.4 years; $p < 0.001$) (Table 1). There was no difference in provider race/ethnicity distribution by arm. The average panel size (per provider) was 1316.

Bivariate Analyses

ATC arm providers made 161 referrals to ATC in year 1 and 126 referrals in year 2 (Table 2) and prescribed more AUD medications than Usual Care providers in year 2 (mean (standard error) = 0.50 (0.09) vs. 0.45 (0.10) scripts on average). UC providers made more referrals to specialty treatment in both years (3.14 (0.28) vs. 2.34 (0.20) in year 1 and 3.24 (0.27) vs. 2.43 (0.21) in year 2). All prescriptions written by ATC pharmacists and/or PCPs in the ATC arm attributed to ATC arm totals.

Multivariable Analyses

The ZIP models accounting for clustering of providers within clinics failed to converge despite trying various starting values for the parameters as is typically done for these analyses, as well as increasing the maximum number of iterations to over 5000. Therefore, the ZIP model was estimated without adjusting for clustering of providers within clinics. We first examined the direct effect of treatment (Table 3), which did not include any covariates, on the outcomes (number of AUD medication orders, and number of referrals per provider). The results showed that the ATC providers were less likely to have no orders (odds ratio (OR); 95% confidence interval (CI) = 0.48; 0.26, 0.88)), i.e., almost

Table 2 Bivariate Distribution of Implementation Outcomes in Eligible Sample[a]

Year 1: September 2021 through August 2022					
Implementation outcome	**ATC arm** (*N = 204*)		**Control** (*N = 132*)		***p*-value**
	N	Mean (SE)	N	Mean (SE)	
AUD medication orders*	103	0.50 (0.09)	60	0.45 (0.10)	0.723
Number of referrals per provider					
Referrals to specialty treatment	477	2.34 (0.20)	415	3.14 (0.28)	0.018
Referrals to ATC	161	0.79 (0.14)	1	0.01 (0.01)	< 0.001
Year 2: September 2022 through August 2023					
Implementation outcome	**ATC arm** (*N = 204*)		**Control** (*N = 132*)		***p*-value**
	N	Mean (SE)	*N*	Mean (SE)	
AUD medication orders*	169	0.83 (0.13)	91	0.70 (0.14)	0.471
Number of referrals per provider					
Referrals to specialty treatment	495	2.43 (0.21)	428	3.24 (0.27)	0.017
Referrals to ATC	126	0.62 (0.09	4	0.03 (0.02)	< 0.001

[a]Eligible patients screened positive for unhealthy alcohol use, received a brief intervention as documented by ICD-10 alcohol counseling Z-code, received an AUD diagnosis, and/or received a referral for an alcohol problem. *ATC*, Addiction Telemedicine Consultation

*AUD medications included acamprosate, disulfiram, and naltrexone

Table 3 Unadjusted Zero-Inflated Poisson Model of Implementation Outcomes

Year 1: September 2021 through August 2022				
Implementation outcome	**Poisson count model Rate ratio**	**p-value**	**Zero inflation model Odds ratio**	**p-value**
Number of medication orders per provider				
AUD medication* orders — ATC arm	1.05	0.79	0.48	0.02
(Reference = usual care)	(0.72, 1.54)		(0.26, 0.88)	
Number of referrals per provider				
Referrals to specialty treatment — ATC arm	1.14	0.06	0.73	0.47
(Reference = usual care)	(1.00, 1.31)		(0.31, 1.72)	
Year 2: September 2022 through August 2023				
Implementation outcome	**Poisson count model Rate ratio**	***p*-value**	**Zero inflation model Odds ratio**	***p*-value**
Number of Medication Orders per Provider				
AUD medication* orders — ATC arm	1.64	0.00	0.72	0.24
(Reference = usual care)	(1.23, 2.17)		(0.42, 1.24)	
Number of referrals per provider				
Referrals to specialty treatment — ATC arm	1.23	0.00	1.29	0.58
(Reference = usual care)	(1.07, 1.40)		(0.52, 3.22)	

*AUD medications included acamprosate, disulfiram, and naltrexone; *ATC*, Addiction Telemedicine Consultation

Table 4 Multivariable Zero-Inflated Poisson Model of Implementation Outcomes

Year 1: September 2021 through August 2022				
Implementation outcome	**Poisson count model Rate ratio**	***p*-value**	**Zero inflation model Odds ratio**	***p*-value**
Number of medication orders per provider				
AUD medication* orders — ATC arm	0.74	0.150	0.36	0.008
(Reference = usual care)	(0.49, 1.11)		(0.17, 0.76)	
Number of referrals per provider				
Referrals to specialty treatment — ATC arm	1.11	0.142	0.87	0.780
(Reference = usual care)	(0.97, 1.28)		(0.33, 2.28)	
Year 2: September 2022 through August 2023				
Implementation outcome	**Poisson count model Rate ratio**	**p-value**	**Zero inflation model Odds ratio**	**p-value**
Number of medication orders per provider				
AUD medication* orders — ATC arm	1.66	0.001	1.08	0.830
(Reference = usual care)	(1.23, 2.24)		(0.55, 2.10)	
Number of referrals per provider				
Referrals to specialty treatment — ATC arm	1.15	0.049	1.31	0.603
(Reference = usual care)	(1.00, 1.31)		(0.48, 3.57)	

*AUD medications included acamprosate, disulfiram, and naltrexone; *ATC*, Addiction Telemedicine Consultation

twice as likely (1/0.48 = 2.08) to order AUD medications than UC in year 1 and ordered more AUD medications in year 2 (rate ratio (RR) = 1.64; 95% CI = 1.23, 2.17)); they also made more referrals to specialty addiction treatment (RR = 1.23; 95% CI = 1.07, 1.40). The intraclass correlation for outcomes varied from a low of 0.05 (for referrals) to 0.13 (for AUD medication orders). In models adjusting for provider characteristics (Table 4), these results were maintained and the Akaike information criterion was lower than for models without covariates (627 vs. 698 for medication model and 1624 vs. 1723 for specialty treatment) suggesting better fit. The ATC arm providers were 2.8 times (1/0.36) more likely to order AUD medications in year 1 and had higher rates of AUD medication prescribing (RR = 1.66, 95% CI = 1.23, 2.24) and referrals to specialty treatment in year 2 (RR = 1.15, 95% CI = 1.00, 1.31) (Table 4).

As a sensitivity analysis, we examined a single-equation Poisson model accounting for clustering, but results did not change substantively (Supplemental Tables 1 and 2)).

DISCUSSION

In this pragmatic, randomized controlled trial, we examined the effects of an innovative, primary care–based approach to address the spectrum of alcohol problem severity using clinical pharmacists trained in MI, AUD pharmacotherapy, and referral facilitation. We examined rates of alcohol medication prescribing and referrals to specialty treatment — outcomes that are important because few people who need treatment for alcohol problems receive it.[57] We found that medication prescribing was almost twice as high in the ATC

arm compared to UC, suggesting that the ATC model was effective at increasing the use of pharmacotherapy.

We found a modest difference between the two arms in specialty treatment referrals. Although a direct comparison between the two arms in ATC referrals is moot as these referrals are only relevant for the ATC arm, it is possible that providers viewed ATC as a substitute for referral to specialty treatment, especially for patients for whom multiple referrals to treatment may have been unproductive. Therefore, if given the option, referral to ATC might be a noteworthy care modality concomitant with (or complementary to) specialty referral that may have bearing on patient outcomes.

The most effective methods to increase adoption of alcohol pharmacotherapy remain unclear. Several studies showed limited or no increases following attempts to enhance uptake.[58–60] By contrast, Saitz et al. found that primary care patients with AUD randomized to receive multidisciplinary Chronic Care Management had greater odds of being prescribed alcohol pharmacotherapy than those randomized to UC,[61] and research from the VHA found that care management strategies did increase alcohol pharmacotherapy use.[62,63] Williams and colleagues, who conducted a systematic review of alcohol pharmacotherapy implementation studies, concluded that it was difficult to determine which strategies were most successful, due to significant variation and unclear descriptions of the interventions.[34]

Few studies have examined interventions by pharmacists in the management of primary or specialty care patients with unhealthy alcohol use. One described a VHA program including clinical pharmacists in an interdisciplinary primary care clinic to treat patients with complex clinical profiles, including AUDs; during the brief study period, pharmacists feasibly managed a large number of patients.[64] Studies have described the value of including a pharmacist on an inpatient psychiatric or specialty treatment team to increase AUD pharmacotherapy initiation; for example, a clinic coordinated by pharmacists to aid in care transitions was associated with increased persistence of extended-release naltrexone use after hospital-based initiation.[65,66] A pharmacist embedded in an intensive outpatient treatment[67] program was also found to improve access to appointments for AUD pharmacotherapy. The current study further demonstrates how pharmacists can enhance access to treatment for unhealthy alcohol use.

Clinical pharmacists are licensed to initiate and manage medications under collaborative practice agreements across most of the USA,[68] and are increasingly being incorporated into primary care teams to increase physician efficiency and reduce burnout.[69] They successfully manage patients with chronic medical conditions such as dyslipidemia,[70] hypertension,[71] diabetes,[72] HIV/AIDS,[73,74] and chronic pain treated with long-term opioid therapy, resulting in improved adherence to treatment guidelines.[75,76] Integrating pharmacists into primary care teams has also been shown to increase PCP[77] and patient satisfaction.[78] In specialty treatment programs, pharmacists have effectively supported individuals receiving medication treatment.[67] ATC is a new model that extends the role of pharmacy in primary care–based treatment of unhealthy alcohol use to a broader population. This approach for treatment of alcohol problems could be incorporated into existing primary care–based, clinical pharmacist-delivered services for managing other chronic conditions, such as depression, diabetes, or hypertension, or alternatively, could be incorporated into pharmacist-delivered models of care for mental health treatment.

While increases in prescribing were significant, specialty care initiation showed no difference. It is possible that low overall ATC uptake may have contributed to these mixed findings. A longer follow-up time combined with the use of PCP champions and/or more PCP reminders in a wider variety of formats (email, instant message) might increase uptake of the intervention and lead to greater and more consistent outcome effects. Qualitative analyses of interviews with PCPs are underway and may shed more light on strategies for increasing uptake as well as improving specialty care referral. Future studies will also examine patient outcomes including prescription fill rates, initiation of specialty treatment, and rates of unhealthy alcohol use, to determine ATC's effect on patient health and wellbeing.

The ATC intervention is generalizable to many types of healthcare settings. Using a centralized model like ATC takes advantage of economies of scale and could expand access to specialty treatment in both public and private settings, from integrated systems like KPNC to rural counties with limited treatment resources. For settings without an EHR-based referral system, referrals to a centralized service could occur by other means such as telephone or secure messaging.[45] While ATC appointments occurred via video or telephone based on patient preference, sites or patients without video capacity could limit their visits to telephone.[79] The factors involved in implementing such a program should not be minimized, however. Health systems considering ATC-type interventions must evaluate pharmacist availability, requirements for training and funding the practitioners, systems for referrals and efficient communication, and availability of escalation to specialty treatment.

Study Limitations and Strengths

We examined ATC as a multi-faceted intervention consisting of access to a pharmacy service, PCP training and reminders, and EHR-based tools, and thus cannot determine the extent to which individual aspects contributed to outcomes. Although we randomized clinics, there were differences in provider characteristics between the two arms; our multivariate analyses helped mitigate this imbalance to some extent. Another potential limitation is contamination between ATC and UC arms, e.g., communication between PCPs in the ATC arm

and their UC colleagues about the services and/or benefits of pharmacotherapy for unhealthy alcohol use. However, block-randomizing by clinic versus by PCPs within clinics reduced the likelihood of this, and only a handful of physicians from outside the intervention arm attempted to access the service via the EHR-based referral system. Increases in prescribing were significant; specialty care initiation showed a marginal difference. It is possible that low overall ATC uptake may have contributed to these mixed findings. A longer follow-up time combined with the use of PCP champions and/or more PCP reminders in a wider variety of formats (email, instant message) might increase uptake of the intervention and lead to greater and more consistent outcome effects. Qualitative analyses of interviews with PCPs are underway and may shed more light on strategies for increasing uptake as well as improving specialty care referral. Future studies will also examine patient outcomes including prescription fill rates, initiation of specialty treatment, and rates of unhealthy alcohol use to determine ATC's effect on patient health and wellbeing.

CONCLUSION

Findings from this pragmatic trial add to the growing literature on primary care–based treatment rates for unhealthy alcohol use and the evolving role of pharmacists. Given low treatment utilization and pharmacotherapy in particular — there is a pressing need for expanded treatment options in primary care. Increasing access in a non-stigmatizing primary care setting via inclusion of team members with the skills and authority to address unhealthy alcohol use, including through pharmacotherapy, could help prevent alcohol problem progression and facilitate AUD recovery. Based on our results, the ATC intervention, delivered by trained pharmacists, may be a promising addition to the primary care armamentarium.

Supplementary Information *The online version contains supplementary material available at https://doi.org/10.1007/s11606-025-09708-y.*

Acknowledgements: *We thank Agatha Hinman for editorial assistance with the manuscript. We thank the KPNC Regional Pharmacy team, including Kerri Butler, PharmD; Lynn DeGuzman, PharmD, BCGP; Stacey Raff, PharmD; Christopher Chang, PharmD, MBA; Sarah Moline, PharmD; and Arzo Razaq, PharmD, for their guidance and collaboration. We also thank all the clinicians and patients of KPNC's Oakland and San Francisco Adult and Family Medicine for their participation in activities related to this study.*

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Funding *This study was supported by National Institute of Alcohol Abuse and Alcoholism grant R01AA028211. The National Institute of Alcohol Abuse and Alcoholism had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.*

Data Availability: *The datasets generated during and/or analyzed in this report are the property of Kaiser Foundation Health Plan, Inc. and are not publicly available due to their potentially identifiable information and Kaiser Permanente Northern California privacy regulations.*

Declarations:

Human Ethics and Consent to Participate: *Not applicable.*

Conflict of Interest: *The authors declare that they do not have a conflict of interest.*

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